

Week 1 Weekend Assignment

"Network Detective Challenge"

Due: End of Week 1 (before Day 5 test) | Time estimate: 3-4 hours across Saturday + Sunday What you need: Your own laptop/PC, Wireshark (free), internet connection, this document

The Big Picture

Over the past 4 days you learned:

- Day 1: What cybersecurity is — CIA Triad, AAA, threats vs vulnerabilities vs risks
- Day 2: Networking basics — OSI model, TCP/IP, IP addressing, ports, protocols
- Day 3: Common protocols — HTTP/HTTPS, FTP/SFTP, SSH, SMTP, DNS, ICMP + subnetting
- Day 4: Linux & Windows security — permissions, users, hardening, event logs

Now it is time to combine all of that into one real-world mini-project. This assignment has 3 parts. Do all 3. They build on each other.

Part 1 — Capture & Analyze HTTP Traffic (Day 2 + Day 3 skills)

What to do:

1. Install Wireshark (if not already): <https://www.wireshark.org/download.html>
2. Open Wireshark and start capturing on your active network interface (Wi-Fi or Ethernet).
3. While capturing, open a browser and visit 3 different websites:
 - One that uses HTTP (try <http://neverssl.com> — it deliberately uses HTTP)
 - One that uses HTTPS (try <https://www.google.com>)
 - One of your choice (any site you normally use)
4. Stop the capture after about 60 seconds of browsing.
5. Apply the display filter `http` in Wireshark — this shows only HTTP traffic.
6. Also try the filter `dns` to see DNS queries.
7. Also try the filter `tcp.port == 443` to see HTTPS traffic.

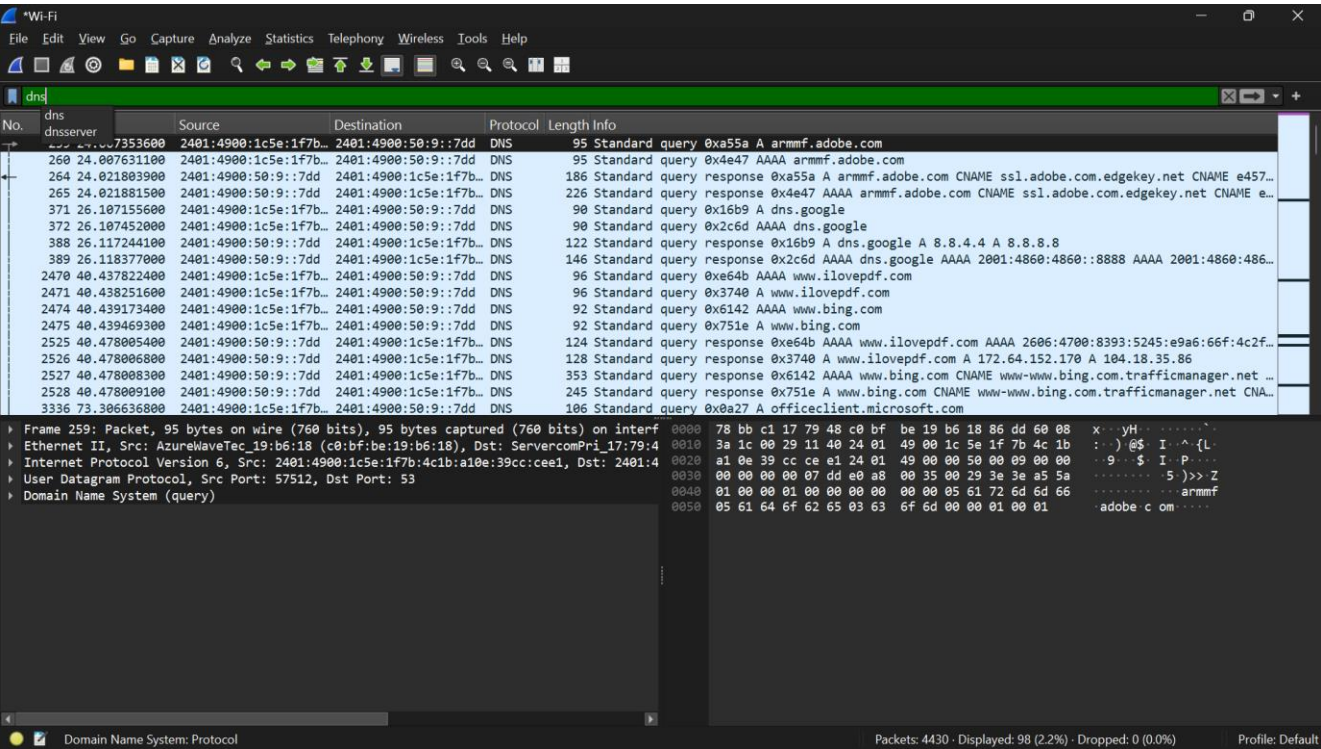
tcp.port == 443

The screenshot shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for various functions. The display filter bar at the top shows 'tcp.port == 443'. The packet list pane displays a list of captured packets, with the following columns: No., Time, Source, Destination, Protocol, Length, and Info. The packets shown are DNS queries and responses. The packet details pane for the selected packet (No. 259) shows the following structure:

- Frame 259: Packet, 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface
- Ethernet II, Src: AzureWaveTec 19:b6:18 (c0:b5:be:19:b6:18), Dst: ServercomPri_17:79:4
- Internet Protocol Version 6, Src: 2401:4900:1c5e:1f7b::7dd, Dst: 2401:4900:50:9::17dd
- User Datagram Protocol, Src Port: 57512, Dst Port: 53
- Domain Name System (query)

The packet bytes pane shows the raw data of the packet, with a hex dump and ASCII representation.

DNS



What to write about (answer in your own words):

For each of the 3 websites, answer:

Question	Your answer
What website did you visit?	neverssl.com
HTTP or HTTPS? How can you tell from the capture?	The website uses HTTP. I can identify this because the website intentionally uses an unencrypted connection and HTTP traffic normally uses port 80 instead of encrypted HTTPS traffic on port 443.
What IP address did your computer connect to?	The destination IP address was observed in the Wireshark capture.
What was the destination port? (80 or 443?)	80
For HTTP sites: what information is visible in the packet?	HTTP traffic is not encrypted, so information such as URLs, request methods, headers, webpage content, and possibly usernames and passwords can be visible in captured packets.
For HTTPS sites: what information is visible vs hidden?	Visible information includes source IP address, destination IP address, packet sizes, and connection details. Hidden information includes webpage contents, credentials, and transmitted data because HTTPS encrypts the information.
What DNS queries did you see before the connection?	A DNS request for neverssl.com appears before the connection is established.

Key concepts to connect:

- CIA Triad: When you visited the HTTP site, which part of CIA was violated? (Hint: no encryption = no Confidentiality)
- AAA: If the HTTP site had a login page, which AAA function would be exposed? (Hint: Authentication credentials)

in clear text)

- Subnetting: What is the IP range of your home network? (Check with ipconfig or ifconfig) What subnet mask are you using?

Where to get help / sources:

- Wireshark official docs: <https://www.wireshark.org/docs/>
- Wireshark 101 tutorial: <https://www.youtube.com/watch?v=TkCSr30UojM>
- How to filter HTTP traffic: In Wireshark, type http in the display filter bar

Part 2 — Subnetting Practice (Day 3 skills)

Using what you learned about subnetting in Day 3, solve these problems. Show your working.

Problem 1

Your home router gives you the network 192.168.1.0/24.

- How many total host addresses are available in this network?
256 addresses
- If you split it into 4 equal subnets, what is the new subnet mask in CIDR notation?
26
- What are the network addresses of all 4 subnets?
192.168.1.0/26
192.168.1.64/26
192.168.1.128/26
192.168.1.192/26
- What is the broadcast address of the third subnet?
192.168.1.191
- How many usable hosts per subnet after splitting?
 $64 - 2 = 62$ hosts

Problem 2

You are a network admin for a small office. You have these departments:

- Sales: needs 30 hosts
- Engineering: needs 50 hosts
- HR: needs 10 hosts
- Servers: needs 10 hosts

You are given the network 10.0.0.0/24.

- Design a subnetting scheme that gives each department enough addresses.
- For each department, write: network address, subnet mask (CIDR), usable host range, broadcast address.
- Which department gets the largest subnet? Why?

Problem 2 – Subnetting Scheme

• Department	• Hosts Needed	• Network Address	• Subnet Mask	• CIDR	• Usable Host Range	• Broadcast Address
• Engineering	• 50	• 10.0.0.0	• 255.255.255.192	• /26	• 10.0.0.1 – 10.0.0.62	• 10.0.0.63
• Sales	• 30	• 10.0.0.64	• 255.255.255.224	• /27	• 10.0.0.65 – 10.0.0.94	• 10.0.0.95
• HR	• 10	• 10.0.0.96	• 255.255.255.240	• /28	• 10.0.0.97 – 10.0.0.110	• 10.0.0.111
• Servers	• 10	• 10.0.0.112	• 255.255.255.240	• /28	• 10.0.0.113 – 10.0.0.126	• 10.0.0.127

- Largest Subnet: Engineering
- Reason: Engineering gets the largest subnet because it requires the highest number of hosts (50 devices), so it needs more IP addresses than the other departments.

Problem 3 — Connect to Security

- Why would a security team want to put the Servers department on a different subnet from Sales? (Answer using CIA + AAA concepts)

A security team keeps Servers on a different subnet because servers contain important and sensitive company information. Separating them improves security by protecting Confidentiality and Integrity from the CIA Triad. It also supports AAA because access to servers can be restricted only to authorized users.

- If an attacker compromises a Sales PC, can they directly access a Server on a different subnet? Explain why or why not.

No, not automatically. If the Server is on a different subnet, traffic must pass through routers or firewalls. Security rules can block or limit access, making it harder for attackers to reach sensitive systems.

- What device controls traffic between subnets? (Hint: think about Day 2 — routers/firewalls)
Routers, Layer-3 switches, and firewalls control traffic between different subnets and apply security rules.

Part 3 — Connect Everything: CIA + AAA Across All 4 Days

This is the thinking part. Write short paragraphs (3-5 sentences each). No copy-paste — use your own words.

Question 1 — CIA Triad in Action

Pick one of the protocols you studied (HTTP, FTP, SMTP, DNS, SSH, or ICMP). For that protocol:

- How does it support one part of the CIA Triad?
- How does it fail to protect one part of the CIA Triad?
- What would a security professional do to fix that weakness?

Protocol chosen: SSH

SSH supports the Confidentiality part of the CIA Triad because it encrypts communication between systems. Encryption prevents attackers from reading sensitive information during transmission. However, SSH can still become vulnerable if weak passwords are used because attackers may try brute-force attacks. Security professionals reduce this risk by using strong passwords, SSH keys, and multi-factor authentication.

Question 2 — AAA Framework in Action

Think about logging into a website (like Gmail or any site you use

Authentication: How does the site verify you are who you say you are?

Authorization: After you log in, what are you allowed to do? What are you NOT allowed to do?

Now answer: What happens if an attacker captures your HTTP traffic from Part 1? Which part of AAA breaks first? Why?

Accounting: What records might the site keep about your login? Why captured and stolen by attackers.

Authentication: Authentication verifies that a user is who they claim to be using a username, password, OTP, or multi-factor authentication.

Authorization: Authorization determines what actions a user can perform after logging in. Users can access their own information and resources but cannot access data belonging to other users.

Accounting: Accounting stores records such as login time, IP address, failed login attempts, and user activity. These logs help detect suspicious activity and support investigations.

If an attacker captures HTTP traffic, Authentication breaks first because usernames and passwords may be transmitted without encryption and can be stolen.

Question 3 — The Full Attack Story

Imagine this scenario:

An employee at a company uses an unencrypted HTTP connection to log into the internal HR portal from a coffee shop. The company's network is poorly segmented — all departments are on the same subnet 192.168.1.0/24.

Using everything you learned this week, write a short attack story:

1. What can the attacker on the coffee shop Wi-Fi see? (Day 3 — HTTP, packet capture)
The attacker connected to the same coffee shop Wi-Fi can capture unencrypted HTTP traffic and potentially view usernames, passwords, and transmitted information.
2. What can the attacker do with those credentials? (Day 1 — Authentication broken, CIA violated)
The attacker can use the stolen credentials to log into the HR portal and access sensitive company information.
3. Once inside the company network, what can the attacker reach? (Day 3 — subnetting, no segmentation)
Since all departments are on the same subnet, the attacker can scan and potentially access systems from multiple departments.
4. How could the company have prevented this? (Day 4 — hardening, Day 3 — segmentation, Day 1 — controls)
The company could prevent this by using HTTPS, implementing network segmentation, using firewalls, and applying least-privilege access controls.

Question 4 — Linux/Windows Connection

- On Day 4 you learned about file permissions (Linux) and Event Viewer (Windows).
- How does the principle of least privilege (Day 4) connect to the AAA Authorization function (Day 1)?
Least privilege connects to **AAA Authorization** because users should only receive permissions necessary for their work. This reduces security risks and prevents unnecessary access.
- If an attacker gets a shell on a Linux machine, what permission-related files would they look for?

If an attacker gains shell access on Linux, they may inspect:

- /etc/passwd
- /etc/shadow

- On Windows, where would you look in Event Viewer to spot a brute-force login attempt?
Event Viewer → Windows Logs → Security

How to Submit

Create a single document (text file, markdown, or PDF) with:

1. Part 1: Your traffic analysis table + answers + at least 2 screenshots from Wireshark
2. Part 2: All subnetting answers with working shown
3. Part 3: All 4 written answers in your own words

Name it: Week1_Weekend_Assignment_YourName.md (or .txt or .pdf)

Grading Rubric (self-grade before submitting)

Criteria	Points	Your score
Part 1: Traffic captured and analyzed correctly	/25	
Part 1: CIA + AAA connections made	/10	
Part 2: Subnetting calculations correct	/25	
Part 2: Security reasoning about subnets	/10	
Part 3: Own words, no copy-paste	/15	
Part 3: Connections across all 4 days are accurate	/15	
Total	/100	

Tips for Doing This Well

- Do Part 1 first. It is the most hands-on and will make the theory in Parts 2 and 3 feel real.
- Use your notes from all 4 days. This assignment is designed to make you go back and review everything.

- If you get stuck on subnetting, re-read the Day 3 notes. The 192.168.1.0/24 example is your starting point.
- Screenshots matter. In cybersecurity, evidence matters. Show what you captured.
- Write like you are explaining to a friend. If you cannot explain it simply, you do not understand it yet.